Bounds and Estimators for the Civilian-to-Combatant Death Ratio from Aggregation of Conflicting Sources

David H. Silver*

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Abstract

Casualty figures in armed conflict are reported by sources with opposing incentives: belligerents inflate enemy combatants killed and deflate civilians; advocacy and intergovernmental bodies push the other way. We treat the civilian-to-combatant death ratio as a partially identified parameter and ask what its value can be when conflicting sources are aggregated. The paper delivers two objects.

(1) Bounds from under-fudgible relations. Under a one-parameter model in which civilian adult men face μ times the death rate of women and children, we give a sharp identified set for the combatant share $q = D_M/D$ as a function of population structure, the demographic composition of the dead, and pre-war combatant stock; intersected with the mechanical bound $D_M \leq M$, this is a closed-form interval. Because the inputs are measured by different, non-aligned institutions, the relation is under-fudgible: moving any one input to rescue a disputed claim forces a compensating, visible move in an independently measured quantity. We define the *contradiction radius* ρ of a claim—the smallest standardised perturbation of an independent input that makes it feasible—turning "we dispute your number" into "name which measurement is wrong, and by how many standard errors."

(2) Confidence coefficients under political bias. Each source is allowed to be Huber ε_i-contaminated; the analyst's political-bias prior on a source enters as ε_i. We give a closed-form first-order sensitivity band that absorbs the assumed contamination, a patternwise worst-case envelope, and a quantile-displacement bound under an explicit posterior-weight condition.

In the Gaza 2023–2026 application—anchored on the Ministry of Health demographic record, whose composition is corroborated by an independent household survey—the identified set for q is $[0, 25\%]$ with no assumption on differential exposure, and $[0, 6\%]$ under exposure ratios calibrated on historical conflicts with known combatant status. The public Israel Defense Forces figure of 17–25k combatants killed ($q \approx$ 24–36%) is rejected under every anchor and every calibrated exposure assumption, with contradiction radii of 8–31 standard errors; the rejection survives correcting the death toll upward and a 5% contamination allowance on every input. Independent methods—demographic decompositions, the military's own leaked internal database of named militant dead, and a fully documented spatial Bayesian model—disagree with one another only *within* the identified set, spanning civilian-to-combatant ratios of roughly 3:1 to 40:1 on direct deaths, while every one of them rejects the public claim of ∼2:1. The same machinery is symmetric: it rejects the polar claim of zero combatant deaths on documentary rather than demographic grounds.

*Independent researcher, Poughkeepsie, NY, USA. Email: `silverdavi@gmail.com`. ORCID: 0000-0002-3071-304X.

1 Introduction

1.1 The aggregation problem

Casualty figures in armed conflict are reported by sources with opposing incentives. Belligerents inflate enemy combatants killed and deflate civilians; advocacy organisations, opposing belligerents, and intergovernmental bodies have symmetric incentives in the other direction. Naive aggregation is therefore biased in a direction that depends on which side the analyst trusts more. The tempting conclusion is that civilian-to-combatant ratios are too politicised to analyse statistically.

We argue for a narrower position. Conflicting source aggregates fail in well-understood ways, but the parameters they purport to measure are jointly constrained by a small number of relations whose inputs are measured by different, non-aligned institutions. An analyst who treats every belligerent source as adversarial can still recover an informative bound by exploiting these constraints; what cannot be recovered without further assumptions is the position of the truth *within* the bound. This separation—bounds from data, position from politics—is the organising principle of the paper. The framework is a *diagnostic*, not an oracle: its purpose is to force any dispute about a casualty claim to name the specific independent measurement being contested, and to quantify how far that measurement would have to be wrong.

A worked example. Suppose a population is 73% women and children ($w = 0.733$) and records of the dead show that 56% of 70,000 dead are women and children ($\omega = 0.560$). If violence were blind to demographics, the dead would mirror the population, so the deficit of women and children among the dead is informative: some of the dead were combatants (all adult men, by assumption), and possibly civilian adult men were more exposed than their population share suggests. If civilian men and women die at *equal* per-capita rates, arithmetic (Section 3) turns these three numbers into a combatant share of $q \approx 25\%$. If civilian adult men die at *twice* the rate of women and children—the kind of differential found when the same method is calibrated on historical conflicts where the answer is known (Frost, 2026)—the same arithmetic gives $q \approx 6\%$. The interval $[\sim 6\%, \sim 25\%]$ is what the data alone deliver; a claimed q of 30% is outside it for *every* admissible exposure ratio and can only be rescued by asserting that the census or the death records are wrong, by an amount the framework computes in standard-error units. That amount is the claim's *contradiction radius.*

The closest precedents are Spagat et al. (2009) on the contestation of war-death estimates, Lum et al. (2013) on multiple systems estimation under reporting failure, Radford et al. (2023) on Bayesian inference of conflict losses with reporting biases, Vesco et al. (2026) on probabilistic UCDP underreporting, and Gómez-Ugarte et al. (2025) on uncertainty quantification for Gaza mortality. Methodologically we draw on partial identification (Manski, 2003, 2007; Molinari, 2020; Kline and Tamer, 2023) and robust Bayesian analysis (Huber, 1964, 1981; Berger and Berliner, 1986). On Gaza specifically, complementary demographic decompositions of the identified dead have been developed by Cockerill (2024) and Frost (2026), and we engage with both.

1.2 Two deliverables

Bounds. Where can the civilian-to-combatant ratio possibly lie, given a set of independently measured demographic and manpower facts? The answer is a sharp identified set in the sense of Manski's partial-identification programme (Manski, 2003; Tamer, 2010; Romano and Shaikh, 2010). If the identified set is narrow, no source-aggregation rule can pull the truth outside it. The width is determined by the data, not by the analyst's politics.

Confidence coefficients. How much can the analyst's political-bias prior on each source widen this set? Each source is Huber ε-contaminated (Huber, 1981; Berger and Berliner, 1986): it is true with probability $1-\varepsilon$ and adversarial with probability ε. A bias-robust sensitivity band inflates the face-value interval by $\sum_i \varepsilon_i R_i$ to first order in ε, where R_i is the diameter of the identified set when source i is removed; conditional on a realised contamination of source i, the worst-case displacement is R_i itself. Setting $\varepsilon_i \approx 1$ for belligerent claims and $\varepsilon_i \approx 0.05$ for census-grade sources gives an interval honest about which inputs are doing the work.

The two objects are complementary: the first is information that survives any reasonable politics; the second is the rate at which uncertainty grows as the analyst dials in distrust of specific sources.

1.3 The under-fudgible mechanism

The bounds are not vacuous because of demographic arithmetic. Consider any claimed decomposition of the dead into combatants and civilians. The claim must simultaneously respect (i) the demographic composition of the identified dead, (ii) the demographic composition of the pre-war population, (iii) the size of the armed group's pre-war stock, and (iv) the plausible range of the civilian adult-male exposure differential. These quantities are measured by *different* institutions: a national or UN census produces (ii); a health ministry, the OHCHR, or a household survey produces (i); independent military-balance assessments produce (iii); and (iv) can be calibrated on historical conflicts where combatant status was recorded (Frost, 2026). They are not equally manipulable, and—critically—they cannot all be moved at once without the moves being visible.

We call a relation between conflict statistics *under-fudgible* when its inputs are reported by non-aligned institutions (or have uncontroversial physical bounds), so that adjusting any single input to rescue a disputed claim forces a measurable deviation in at least one independently measured quantity. This is a heuristic property of a measurement system, not a formal axiom; what is formal is its consequence, the contradiction radius of Section 4, which quantifies exactly how large the forced deviation is for any specific claim. The trigger is *joint inconsistency*, not the size of the claim: the same machinery rejects a claimed combatant share that is too high and one that is too low, against the same feasible set. This symmetry matters. The diagnostic responds to the demographic content of a claim, not to its political direction, and Section 6 applies it in both directions.

1.4 Contributions and roadmap

The paper delivers, in order of importance:

(i) *A sharp identified set* for the combatant share $q = D_M/D$ under bounded differential exposure of civilian adult men, intersected with the manpower bound $D_M \leq M$ (Section 3). The set is a closed-form interval computable from four numbers.

(ii) *The contradiction radius* ρ: a feasibility test for any disputed claim, stated as a distance-to-feasibility optimisation in standard-error units (Section 4). It converts "we dispute your number" into "name which independent input is wrong, and by how many standard errors."

(iii) *A bias-robust sensitivity band* under Huber ε-contamination of each source: a pattern-wise worst-case envelope, a first-order band for the interval endpoints, and a quantile-displacement bound under an explicit posterior-weight condition (Section 5).

(iv) *A precision-weighted aggregation rule* for multiple demographic sources, with an explicit caveat for selection-biased sources (Section 2.1).

(v) *A Gaza 2023–2026 application* (Section 6): the identified set, the contradiction radius of the public IDF combatant-death claim under every anchor and exposure assumption, a sensitivity grid over all contested inputs, and a comparison against every independent method known to us—which all reject the claim while disagreeing only within the identified set. A fully documented spatial Bayesian model (Appendix B) illustrates where a specific set of modelling assumptions lands *within* the set.

(vi) *An 81-conflict overview* (Section 7) mapping where the diagnostic can and cannot be applied sharply, with per-conflict tables in the supplement.

Proofs are collected in Appendix A. The method does not assign moral responsibility, estimate indirect deaths, or replace forensic work.

2 Setup

A conflict produces $D \in \mathbb{N}$ deaths in $[t_0, t_1]$ in a population of size N. Each death is a combatant ($C = 1$) or civilian ($C = 0$); write $D_M = \sum_i \mathbf{1}\{C_i{=}1\}$, $D_C = D - D_M$, and the targets

$$q = D_M/D \in [0,1], \qquad r = D_C/D_M = (1-q)/q.$$

Each death has a demographic class $k \in \mathcal{K} = \{\text{child}, \text{adult-female}, \text{adult-male}\}$. To keep the two central quantities visually distinct we reserve the *Latin* letter w for the population and the *Greek* letter ω for the dead:

$$\begin{aligned} w &= \text{women-and-children share of the } \underline{\text{population}}, \\ \omega &= \text{women-and-children share of the } \underline{\text{dead}}. \end{aligned}$$

Formally, $\omega_k = D_k/D$ is the share of class k among the dead and p_k its share in the population; AM denotes adult males (18 and over), with population share $a = p_{\mathrm{AM}} = 1-w$; $\omega = \sum_{k\neq\mathrm{AM}} \omega_k$. Let M be the pre-war combatant stock and $f = M/N$. We take combatants to be in AM; female and minor combatants enter as a small slack $\bar{\eta} \le 0.03$ (Remark 3).

Table 1. Core notation and measurement channels. The distinction between w (population) and ω (dead) is load-bearing throughout: w comes from a census taken before or independently of the fighting, ω from records of the dead.

Symbol	Meaning	Role	Typical source
D	total deaths in window	input	ministry of health, OCHA, UCDP, survey
D_M, D_C	combatant, civilian deaths	target	decomposition of D
$q = D_M/D$	combatant share of dead	estimand	—
w	women+children share *of population*	input	census, UN Population Division
$a = 1-w$	adult-male (18+) share *of population*	input	census / age pyramid
ω	women+children share *of the dead*	input	identified-deaths records (e.g. MoH)
M	combatant stock at t_0	input	IISS / RUSI / CSIS / intelligence
$[\mu_{\mathrm{lo}}, \mu_{\mathrm{hi}}]$	admissible exposure-ratio range	assumption	calibrated from historical conflicts
ρ	contradiction radius of a claim	output	SE-units to feasibility; see §4

The last row anticipates the paper's diagnostic. The *contradiction radius* ρ of a disputed claim is the smallest number of measured standard errors by which at least one independent input (ω, w, or M) would have to move for the claim to become consistent with all the others; $\rho = 0$ means consistent, $\rho = 3$ means some independently measured quantity must be wrong by three standard errors, and values in the double digits mean the claim requires an input to be wrong by an amount that measurement error cannot plausibly produce. Section 4 makes this precise.

Sources. We use five channels: an independent population census (p_k, N); demographic records of the dead returning sample shares $\hat{\omega}_k$; aggregate death totals $\hat{D}$; belligerent combatant-death claims $\hat{D}_M$; and independent manpower estimates $\hat{M}$. Each source i is allowed to be ε_i-contaminated in Huber's sense (Huber, 1964; Berger and Berliner, 1986): it draws from the truth with probability $1 - \varepsilon_i$ and from an arbitrary adversary with probability ε_i.

2.1 Source aggregation

When several independent sources $i = 1, \ldots, m$ report the same scalar parameter θ with reported point estimates $\hat{\theta}_i$ and standard errors $\hat{\sigma}_i$, the natural *precision-weighted* aggregate is

$$\hat{\theta}_{\text{agg}} = \frac{\sum_i \hat{\theta}_i / \hat{\sigma}_i^2}{\sum_i 1/\hat{\sigma}_i^2}, \qquad \hat{\sigma}_{\text{agg}}^2 = \Big(\sum_i 1/\hat{\sigma}_i^2\Big)^{-1}. \tag{1}$$

Proposition 2.1 (Aggregation under conflicting sources)**.** *If sources are independent and unbiased, with variances treated as known (with estimated $\hat{\sigma}_i^2$ the statement holds conditionally on the variance estimates and asymptotically), $\hat{\theta}_{\text{agg}}$ in* (1) *is the minimum-variance unbiased linear estimator of θ. If source i has bias b_i, the aggregate has bias $(\sum_i b_i/\hat{\sigma}_i^2)/(\sum_i 1/\hat{\sigma}_i^2)$. Allowing ε_i-contamination (Section 5) in which the contaminated report is displaced from the truth by at most R_i, the worst-case bias of $\hat{\theta}_{\text{agg}}$ is at most $\sum_i \varepsilon_i R_i \cdot \hat{\sigma}_{\text{agg}}^2/\hat{\sigma}_i^2$.*

Proof in Appendix A.

Precision weighting presupposes that the sources measure the *same* estimand. When one source is subject to a known selection mechanism—as with the OHCHR verification sample in the Gaza application, which is heavily weighted toward residential-strike deaths (Spagat and Cockerill, 2024)—the honest treatment is not to blend it into the aggregate but to use it as a stratum-specific measurement and anchor the analysis on the source whose selection is best understood. Section 6 applies this rule: the Ministry of Health record, whose demographic composition is corroborated by an independent household survey (Spagat et al., 2026a), is the primary anchor, and results are reported for the alternative anchors so the reader can see exactly how anchor choice propagates.

3 Bounds: partial identification of q

This section delivers a sharp identified set for q that any source-aggregation rule must respect. We begin with a benchmark assumption, then weaken it. Proofs are in Appendix A.

Assumption 3.1 (Exchangeable collateral)**.** Civilians of all demographic classes face equal per-capita death hazard.

Under Assumption 3.1 the combatant share is point-identified by a single rearrangement: civilian deaths split between the women-and-children class and the adult-male class in proportion to their civilian population shares, so

$$q = 1 - \frac{\omega(1-f)}{w}. \tag{2}$$

The intuition is direct: ω falls short of the population share w only to the extent that deaths are diverted to the (all-male) combatant pool.

Remark. Suppose a fraction $\eta \leq \bar{\eta} = 0.03$ of combatant *deaths* is recorded in class W. This is a classification slack: the combatant stock itself stays in AM, so the civilian split

$S = w/(w + \mu(a - f))$ is unchanged. The identity becomes $\omega = (1 - q)S + \eta q$, the true share is $q_{\text{true}} = (S - \omega)/(S - \eta)$, and the no-slack value $q_{\eta=0} = (S - \omega)/S$ understates it by exactly

$$q_{\text{true}} - q_{\eta=0} \;=\; \frac{\eta\, q_{\text{true}}}{S} \;=\; \frac{\eta\, q_{\eta=0}}{S - \eta} \;\geq\; 0 \qquad (S > \eta,\; q_{\text{true}} \geq 0).$$

At the values in Section 6, where $S > \bar{\eta}$ throughout ($S \geq 0.597$ for $\mu \leq 2$) and $q_{\eta=0} \leq 0.26$, a loose uniform bound over $\mu \in [1, 2]$ is $\bar{\eta} \cdot 0.26/(0.597 - \bar{\eta}) \approx 1.4$ percentage points; the exact understatement is smaller and falls with μ (1.05 points at $\mu = 1$, 0.33 at $\mu = 2$), since $q_{\eta=0}$ shrinks faster than S. Readers who want this slack should widen the *upper* endpoints below by $\bar{\eta}\, q_{\eta=0}/(S - \bar{\eta})$; we otherwise suppress it.

Assumption 3.1 is strong: civilian adult men typically face higher per-capita exposure than women and children due to outside work, food queues, rescue activity, and circumstantial proximity to combat. We weaken it to a one-parameter family.

Assumption 3.2 (μ-bounded differential exposure)**.** There exists $\mu \in [\mu_{\text{lo}}, \mu_{\text{hi}}]$ such that, conditional on being civilian, class AM has per-capita death rate μ times that of class W.

Theorem 3.3 (Sharp identified set)**.** *Suppose $f \leq a$ and Assumption 3.2 holds. Given (ω, w, a, f) and the exposure range $[\mu_{\text{lo}}, \mu_{\text{hi}}]$, the identified set for q is the image of the monotone-decreasing curve*

$$q(\mu) = 1 - \omega\, \frac{w + \mu(a - f)}{w}, \qquad \mu \in [\mu_{\text{lo}}, \mu_{\text{hi}}], \tag{3}$$

giving the sharp interval $[q(\mu_{\text{hi}}),\, q(\mu_{\text{lo}})] \cap [0, 1]$.

Corollary 3.4 (Manpower bound)**.** *$q \leq M/D$, so whenever the endpoints below are ordered, the joint feasible interval is*

$$q \in \big[\max\{0,\, q(\mu_{\text{hi}})\},\; \min\{1,\, q(\mu_{\text{lo}}),\, M/D\}\big];$$

if the lower endpoint exceeds the upper the set is empty and the inputs are jointly infeasible—the event detected by a positive contradiction radius in Section 4.

Calibrating the exposure range. The exposure ratio μ is the framework's only behavioural parameter, and it can be disciplined empirically rather than stipulated. Frost (2026) identifies the civilian male exposure differential from the male-to-female death ratio just *above* the combatant age range—where the combatant share is mechanically near zero—and validates the procedure on historical Gaza conflicts recorded by B'tselem, where combatant status is known: there, the observed male-to-female ratio at the top of the combatant age band is 3.4:1 while the effective differential through the combatant ages is 5:1, so the observable ratio is itself a conservative lower bound. Airstrike-specific data suggest smaller differentials (≈ 1.3; Cockerill, 2024), consistent with indiscriminate mechanisms; ground operations push the differential up. We therefore report the identified set on a grid $\mu \in \{1, 1.5, 2, 2.5, 3.5, 5\}$ throughout, treating $[2, 3.5]$ as the empirically best-supported range for a mixed air-and-ground campaign and the extreme values as stress tests. Larger μ *lowers* the implied combatant share; the reader who disputes a high-μ calibration is pushed toward *higher* q, and the sensitivity grid (Table 3) makes the entire mapping explicit.

Plug-in estimator and standard error. The sample analogue of the upper endpoint is $\hat{q} = 1 - \hat{\omega}(1 - \hat{f})/\hat{w}$, with delta-method variance

$$\widehat{\mathrm{Var}}(\hat{q}) \approx \frac{\hat{\omega}^2}{\hat{w}^2}\widehat{\mathrm{Var}}(\hat{f}) + \frac{(1 - \hat{f})^2}{\hat{w}^2}\widehat{\mathrm{Var}}(\hat{\omega}) + \frac{\hat{\omega}^2(1 - \hat{f})^2}{\hat{w}^4}\widehat{\mathrm{Var}}(\hat{w}).$$

4 Bounds: contradiction radius

Theorem 3.3 returns an interval; a public claim returns a point. The diagnostic measures their distance after standardising each input by its measurement uncertainty.

Definition 4.1 (Feasible set)**.** For inputs $x = (\omega, w, a, f, M, D)$ and exposure range $[\mu_{\text{lo}}, \mu_{\text{hi}}]$,

$$\mathcal{F}(x) = \Big\{ q \in [0,1] : \exists\, \mu \in [\mu_{\text{lo}}, \mu_{\text{hi}}] \text{ with } q = 1 - \omega^{\frac{w+\mu(a-f)}{w}} \text{ and } qD \le M \Big\}. \quad (4)$$

A claim $\hat{D}_M^{(i)}$ converts to $q^{(i)} = \hat{D}_M^{(i)}/\hat{D}$ and is feasible iff $q^{(i)} \in \mathcal{F}(\hat{x})$.

Definition 4.2 (Contradiction radius)**.** Let $\hat{x} = (\hat{\omega}, \hat{w}, \hat{a}, \hat{f}, \hat{M}, \hat{D})$ with measured standard errors $\sigma_\omega, \sigma_w, \sigma_M$. The contradiction radius of claim $q^{(i)}$ is

$$\rho^{(i)} = \inf_{\omega', w', M'} \max\Big\{ \tfrac{|\omega' - \hat{\omega}|}{\sigma_\omega}, \ \tfrac{|w' - \hat{w}|}{\sigma_w}, \ \tfrac{(M' - \hat{M})_+}{\sigma_M} \Big\} \quad \text{s.t.} \quad q^{(i)} \in \mathcal{F}((\omega', w', \hat{a}, \hat{f}, M', \hat{D})). \quad (5)$$

Theorem 4.3 (Feasibility and contradiction)**.** *With $(\hat{a}, \hat{f}, \hat{D})$ and the exposure range fixed, a claim is consistent with the independent inputs iff $\rho^{(i)} = 0$. If $\rho^{(i)} > 0$, every rationalisation (ω', w', M') makes at least one* penalised *displacement—$|\omega' - \hat{\omega}|/\sigma_\omega$, $|w' - \hat{w}|/\sigma_w$, or $(M' - \hat{M})_+/\sigma_M$—of at least $\rho^{(i)}$ standard-error units. (Lowering M is unpenalised because it can only shrink the feasible set.)*

Interpretation. The contradiction radius answers the question a motivated critic must eventually face: *which measurement, exactly, do you say is wrong, and by how much?* A claim with $\rho = 0$ is compatible with all independent inputs; nothing more can be said against it on this evidence. A claim with $\rho = 2$ requires some input to be off by two standard errors—uncomfortable but conceivable. A claim with $\rho = 20$ requires an independently measured quantity to be wrong by twenty standard errors, an error that measurement noise cannot produce; defending such a claim requires asserting institutional fabrication, and the radius names the institutions and the magnitudes. Because ρ standardises by *measured* uncertainty, it is unit-free and comparable across conflicts and across claims. It is also symmetric: an implausibly *low* combatant claim generates a positive radius the same way an implausibly high one does. What the radius is not: it is not a posterior probability, it does not weigh how *many* inputs must move (only the largest single standardised move), and it inherits any systematic—as opposed to sampling—error in the inputs, which is why Section 6 reports it under every anchor choice.

Computation. Because $q(\mu)$ is monotone, the boundary of $\mathcal{F}$ is attained on one of three axes: the demographic axis (ω moves), the population-composition axis (w), or the manpower axis (M/D). This is the contradiction triangle of Figure 1; (5) is a low-dimensional convex programme.

Diagnostic procedure.

D1. Fix the analysis window and total deaths D; carry the uncertainty in D as a sensitivity dimension, not a constant.

D2. Estimate (w, a) from the population baseline and ω from identified-deaths records, checking each candidate ω source for selection mechanisms before use.

D3. Calibrate the exposure range $[\mu_{\text{lo}}, \mu_{\text{hi}}]$ from historical conflicts with known combatant status, and take an independent upper bound on M.

D4. Compute $\mathcal{F}(\hat{x})$ from (4).

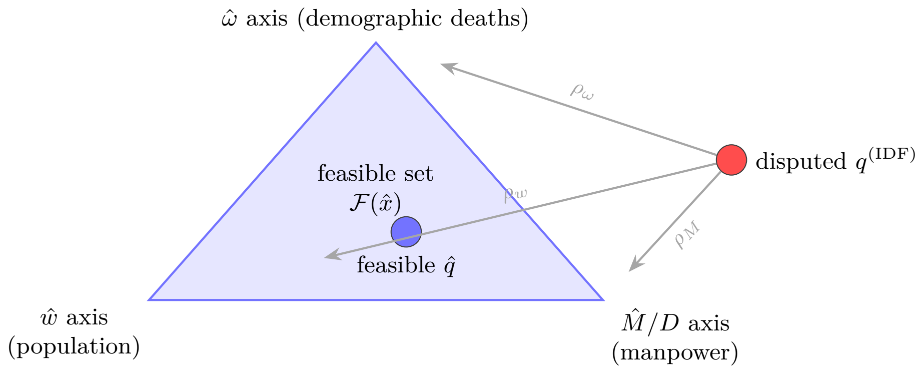

Figure 1. Contradiction triangle (conceptual schematic). The feasible set $\mathcal{F}$ is bounded by three independently measured "axes"; the contradiction radius $\rho = \min(\rho_\omega, \rho_w, \rho_M)$ is the shortest standardised perturbation from a disputed claim into $\mathcal{F}$. Each axis is owned by a different institution, which is what makes the relation under-fudgible: rescuing a claim on one axis leaves fingerprints on that axis's measurement.

D5. For every disputed $\hat{D}_M^{(i)}$, report $q^{(i)} = \hat{D}_M^{(i)}/D$, the membership $q^{(i)} \in \mathcal{F}$, and $\rho^{(i)}$, under each anchor and each D scenario.

The test is symmetric: it rejects "too many combatants" and "zero combatants" against the same set.

5 Confidence coefficients under political bias

The bounds in Sections 3–4 treat each measured input as honest. We now allow the analyst to dial in distrust of specific sources. Political bias enters the confidence statement through Huber contamination: each source i writes its likelihood as $(1-\varepsilon_i)L_i + \varepsilon_i Q_i$, with Q_i adversarial in some class $\mathcal{Q}$ (Huber, 1964, 1981; Berger and Berliner, 1986; Gagnon, 2023). The contamination parameter $\varepsilon_i \in [0,1]$ is the analyst's prior probability that source i is producing a politically motivated reading rather than a measurement of the true distribution.

Proposition 5.1 (Bias-robust sensitivity band)**.** *Let θ be a scalar estimand with $1-\alpha$ posterior credible interval $I_0 = [L_0, U_0]$ under face-value sources, with the posterior supported on the identified set implied by the sources used. Suppose each source i is independently contaminated with probability ε_i, and removing the sources in a set T leaves an identified set of diameter R_T (write $R_i := R_{\{i\}}$).*

(i) (Patternwise envelope.) *If the realised contamination pattern is T, every* support-respecting *$1-\alpha$ credible interval—one whose endpoints lie in the posterior support, e.g. the equal-tailed or HPD interval—obtainable by adversarial replacement of the sources in T lies within $[L_0 - R_T,\ U_0 + R_T]$. (An arbitrary credible interval can be widened without limit while retaining $1-\alpha$ coverage, so no finite envelope constrains that class.)*

(ii) (Expected displacement.) *Averaging over the contamination pattern, the expected endpoint displacement is at most $\sum_i \varepsilon_i R_i$ plus higher-order terms in ε involving the multi-removal diameters R_T, $|T| \geq 2$; when at most one source can be contaminated, the bound $\sum_i \varepsilon_i R_i$ is exact. The band*

$$I_\varepsilon = \left[L_0 - \textstyle\sum_i \varepsilon_i R_i,\ U_0 + \sum_i \varepsilon_i R_i\right] \tag{6}$$

is therefore a first-order sensitivity band for the interval endpoints—not a uniform envelope over realised contaminated intervals, whose endpoints can move by up to R_T on the (probability $\leq \varepsilon_i$) contamination events.

(iii) (Quantile displacement.) *If additionally the posterior mixture weight on contaminated components is at most* $\tau := 1 - \prod_i (1 - \varepsilon_i)$*—which holds when the component marginal likelihoods are comparable, and must otherwise be checked, since posterior weights are proportional to prior pattern probabilities* times *component marginal likelihoods—and the face-value posterior density exceeds* $m > 0$ *on the segment between the clean and contaminated* u*-quantiles, then*

$$|\hat{Q}_\varepsilon(u) - \hat{Q}_0(u)| \;\le\; m^{-1}\tau, \qquad u \in \{\alpha/2,\, 1-\alpha/2\}.$$

Proof in Appendix A.

Practice. Set $\varepsilon_i \approx 0.05$ for widely-trusted demographic sources (Palestinian Central Bureau of Statistics, health-ministry demographic records corroborated by independent surveys) and $\varepsilon_i \approx 1$ (i.e., adversarial) for belligerent claims; (6) then absorbs the assumed political bias automatically. In the Gaza application the removal diameters are $R_{\text{demo}} = 0.64$ (dropping the demographic records leaves only the manpower bound $q \le M/D$), $R_{\text{census}} = 0.06$ (letting w range over a generous global envelope $[0.70, 0.76]$), and $R_M = 0$ (the manpower bound never binds), so $\varepsilon = 0.05$ inflates the exposure-agnostic upper bound by 3.5 percentage points—from 25.1% to 28.7%—which still rejects the upper end of the disputed claim.

6 Application: Gaza 2023–2026

We instantiate the framework using only inputs whose measurement is not controlled by either belligerent: PCBS population structure (Palestinian Central Bureau of Statistics, 2023–2024); OCHA and Gaza MoH total deaths (United Nations Office for the Coordination of Humanitarian Affairs, 2025–2026); OHCHR and MoH demographic death records (United Nations Office of the High Commissioner for Human Rights, 2024); the Gaza Mortality Survey household study (Spagat et al., 2026a); excess-mortality information (Khatib et al., 2024, 2025); and IISS / RUSI / CSIS combatant-strength estimates (International Institute for Strategic Studies, 2024; Royal United Services Institute, 2024). The IDF claim itself (Israel Defense Forces, 2024–2026) is the object of the test, not an input. All numbers in this section are produced by a validation script in the companion repository that recomputes every bound, radius, and table entry from the raw inputs and fails if any quoted figure disagrees.

Table 2. Diagnostic inputs for Gaza. The disputed quantity D_M is tested against the other rows. AM is males 18 and over, so $a = 1 - w$ exactly.

Input	Value	Channel
w (women+children *of population*)	0.733	PCBS population structure
a (males 18+ *of population*)	0.267	$= 1 - w$
ω (women+children *of the dead*)	0.560 (MoH record; primary anchor); 0.693 (OHCHR verified sample)	identified deaths, corroborated by GMS; OHCHR = residential-strike stratum
f	0.0157–0.0269 (0.020 central)	Hamas/PIJ pre-war manpower / population
D	70k (MoH) / 94k (GMS-corrected) / 106k (+missing)	direct deaths, late 2025
μ	grid $\{1, 1.5, 2, 2.5, 3.5, 5\}$; calibrated $[2, 3.5]$	Frost (2026); Cockerill (2024)
M	$\le$ 45,000	IISS / RUSI / CSIS
claim	17–25k combatants killed	IDF public statements

Anchor choice. The two demographic records of the dead measure different things. The OHCHR sample ($n = 8{,}119$) applies a strict three-source verification rule that over-represents deaths in residential strikes, where women and children predominate (Spagat and Cockerill, 2024); it is best read as a stratum-specific measurement, not an estimate of the overall ω. The MoH full record ($n \approx 70{,}000$) covers all recorded deaths, and its demographic composition is independently corroborated by the Gaza Mortality Survey, a household probability survey that also quantifies the MoH undercount at roughly 35% (Spagat et al., 2026a); the Israeli military has itself since accepted the MoH cumulative toll (Kubovich and Hasson, 2026). We therefore anchor on the MoH record ($\omega = 0.560$, conservative $\sigma_\omega = 0.01$) and report the OHCHR anchor—and a geometric blend of the two—as sensitivities rather than blending them into a single aggregate. Because $\omega_{\text{MoH}} < \omega_{\text{OHCHR}}$, this choice is *favourable* to high-combatant claims: every rejection below survives, a fortiori, under the alternative anchors.

The survey's representativeness has been contested: DellaPergola and Zlochin (2026) identify interviewer-level heterogeneity and protocol deviations and argue the population-level extrapolation is unreliable; Spagat et al. (2026b) respond that the conclusions survive excluding the contested teams (the violent-death estimate falls from 75,200 to 59,200 under maximal exclusion, with the demographic composition intact). For present purposes the dispute is instructive precisely because it does not touch the bounds. It concerns the *level* of the death toll, an input our identified set does not use: $q(\mu)$ depends on (ω, w, a, f) only, and the demographic composition—the one quantity the framework takes from the survey—is stable across every specification in the exchange, including the critics' own exclusions and the extreme all-missing-dead scenario (the survey's women-children-elderly share of violent deaths moves from 56.2% only to 52.2%; note this classification includes people over 64 and so is not identical to our ω class, but it is the composition measure the exchange contests). The undercount factor enters solely through the D-sensitivity dimension of Step 3 ($D = 70$–106k), and only its upward corrections matter there. Downward revisions of the toll require no new scenario, because the direction of the challenge is self-defeating for the claim it might appear to assist: if the critics were right and the true toll were *lower*, the claimed combatant counts would convert to *higher* shares $q^{(i)} = \hat{D}_M^{(i)}/D$, moving the disputed claim further outside the feasible set while leaving the set itself unchanged. This is the under-fudgible structure at work: contesting one measurement re-routes, but does not relieve, the burden of the contradiction.

Step 1: rejecting uniform random violence. The naive null "violence is uniform across the population" predicts $\mathbb{E}\,\omega = w = 73.3\%$. Even the OHCHR sample ($\hat{\omega} = 69.3\%$) rejects this at $z = -8.1$ ($p \approx 1.9 \times 10^{-16}$); the MoH record ($\hat{\omega} = 56.0\%$) rejects it overwhelmingly. Adult males are overrepresented among the dead—necessarily, since the universe contains both combatants and structurally more-exposed civilian men. Rejecting uniformity is necessary but not sufficient to rationalise any *specific* combatant share; the question is how the adult-male excess splits between the two explanations, and that is precisely what μ indexes.

Step 2: the identified set. On the MoH anchor, Theorem 3.3 gives

$$q(1) = 25.1\%, \qquad q(1.5) = 15.7\%, \qquad q(2) = 6.3\%, \qquad q(\mu) \le 0 \text{ for } \mu \ge 2.3.$$

The exposure-agnostic set (any $\mu \ge 1$; the upper endpoint is $q(1)$ and the lower is clipped at zero) is $q \in [0,\, 25.1\%]$; the empirically calibrated set ($\mu \in [2, 3.5]$, Frost, 2026) is $q \in [0,\, 6.3\%]$. The manpower bound $M/D = 0.64$ never binds. Table 3 reports the full anchor$\times\mu$ grid; the OHCHR anchor caps q at 7.3% already at $\mu = 1$.

Step 3: testing the IDF claim. Over the MoH-confirmed denominator $D = 70$k, the claimed 17–25k combatants convert to $q^{\text{IDF}} = 24.3\%$–35.7%, i.e. a civilian-to-combatant ratio

Table 3. Sensitivity of the identified-set upper endpoint. Each cell is $q(\mu) = 1 - \omega\,(w + \mu(a - f))/w$ evaluated at $w = 0.733$, $a = 0.267$, $f = 0.020$: the largest combatant share consistent with anchor ω if civilian adult men die at exactly μ times the per-capita rate of women and children. The identified set for exposure range $[\mu_{\text{lo}}, \mu_{\text{hi}}]$ is $[q(\mu_{\text{hi}}),\, q(\mu_{\text{lo}})] \cap [0, 1]$. Dashes: $q(\mu) \leq 0$, i.e. the anchor is consistent with zero combatants at that exposure ratio. Empirically calibrated exposure ratios are $\mu \gtrsim 2$ (Frost, 2026; Cockerill, 2024).

Anchor ω (share of dead)	μ=1	μ=1.5	μ=2	μ=2.5	μ=3.5	μ=5
MoH (GMS-validated), $\omega = 0.560$	25.1%	15.7%	6.3%	–	–	–
Blend (geometric), $\omega = 0.620$	17.1%	6.7%	–	–	–	–
OHCHR (residential-strike stratum), $\omega = 0.693$	7.3%	–	–	–	–	–

of roughly 2:1–3:1. The contradiction radius of Definition 4.2—which grants the claim the most favourable μ in the admissible range—splits cleanly across the two endpoints:

- *The* 25*k endpoint is infeasible outright.* Rationalising $q = 35.7\%$ on the MoH anchor requires $\mu = 0.44$: civilian adult men would have to die at less than half the per-capita rate of women and children, which no mechanism in this conflict supports. On the demographic axis the radius is $\rho_\omega \approx 7.9$ standard errors with agnostic exposure and ≈ 17.6 with calibrated exposure (and ≈ 21–31 on the OHCHR anchor).
- *The* 17*k endpoint survives arithmetic but not calibration.* At $q = 24.3\%$ the required exposure ratio is $\mu = 1.04$—inside the mathematically admissible range, so $\rho = 0$ under agnostic exposure on the MoH anchor. But $\mu \approx 1$ asserts that civilian men in Gaza were no more exposed than women and children, contradicting the calibrated range $\mu \in [2, 3.5]$ obtained from historical Gaza conflicts where combatant status is recorded (Frost, 2026). Under calibrated exposure the radius is $\rho_\omega \approx 10.8$ standard errors.

The rejection is robust to the denominator: correcting the MoH undercount ($D = 94$k) and adding the missing ($D = 106$k) lowers q^{IDF} to 16–24%, but the required exposure ratios ($\mu \leq 1.5$ for 17k, $\mu \leq 1.1$ for 25k) remain below the calibrated range. (Applying the recorded ω to the corrected totals assumes the unrecorded dead share the recorded demographic mix, which the survey's demographic breakdown supports.) The first-order sensitivity band of Proposition 5.1, with $\varepsilon = 0.05$ on the census and demographic records, inflates the exposure-agnostic upper endpoint from 25.1% to 28.7%—still below the claim's upper endpoint. (The patternwise worst case is starker but works in the same direction: if the demographic record itself is the contaminated source, the analysis collapses to the manpower bound $q \leq 0.64$, which is exactly why the record's corroboration by an independent household survey carries the weight it does.)

Step 4: the named-combatant cross-check. A joint investigation by *The Guardian*, +972 Magazine, and Local Call (Kierszenbaum et al., 2025) reported that the Israeli Military Intelligence directorate's internal database listed 8,900 named Hamas and PIJ operatives dead or probably dead as of May 2025, against a contemporaneous recorded toll of ∼53,000: $q \approx 16.8\%$, inside the exposure-agnostic identified set and far below the public claim of the same period. We treat this as corroborating, not primary, evidence—the database's coverage and classification rules are not public—but it is notable that the belligerent's own accounting apparatus lands inside the bound its public statements violate.

Step 5: convergence of independent methods. Table 4 is the central empirical exhibit. Methods with different data, assumptions, and failure modes—the identified set of this paper, demographic decompositions (Cockerill, 2024; Frost, 2026), the military's internal database, and a fully specified spatial Bayesian model (Appendix B)—disagree with one another only *within* $q \in [\sim2\%, \sim25\%]$, that is, civilian-to-combatant ratios from roughly 3:1 to 40:1 or more on direct deaths. Every one of them rejects the public claim of $q \geq 24\%$ ($\leq$ 2–3:1). The spatial model, whose priors and structure are documented in Appendix B, posterior-concentrates at $q = 2.5\%$ (95% credible interval $[1.6\%, 3.8\%]$, civilian-to-combatant ratio 43:1 $[30, 64]$); we report it as one model-based point within the identified set, not as the headline, because its position within the set is prior-sensitive in ways the bounds are not.

Table 4. Convergence of independent methods on the Gaza combatant share. Every method that uses independently measured inputs rejects the public IDF figure; they disagree with one another only *within* $q \in [\sim 2\%, \sim 25\%]$ (civilian-to-combatant ratios of roughly 3:1 to 40:1 or more on direct deaths).

Method	Source	Implied q	Note
IDF public claim (17–25k of 70k)	(Israel Defense Forces, 2024–2026)	24.3–35.7%	object of the test
Identified set, MoH anchor, $\mu \geq 1$	this paper	0.0–25.1%	exposure-agnostic
Identified set, MoH anchor, $\mu \in [2, 3.5]$	this paper	0.0–6.3%	Frost-calibrated exposure
Aman named-militant database	(Kierszenbaum et al., 2025)	16.8%	8,900 named dead of 53,000 (May 2025)
Male-bias model, MoH record	(Frost, 2026)	11.1–16.9%	B'tselem-calibrated
Demographic decomposition	(Cockerill, 2024)	9.4–26.3%	range of male-bias assumptions
Spatial Bayesian posterior	this paper, App. B	1.6–3.8%	prior-dependent; one point in the set

Symmetry. The same machinery tests the polar narrative $q = 0$ ("all the dead are civilians"). On the MoH anchor, $q = 0$ requires $\mu = 2.33$—inside the calibrated range—so the demographic axis alone cannot reject it; the arithmetic is compatible with a purely civilian death toll if civilian men were about 2.3 times as exposed. What rejects $q = 0$ is the named-combatant evidence: thousands of individually named militant dead are documented in the military's own database (Kierszenbaum et al., 2025). The diagnostic thus does exactly what it should: it rejects the high claim on demographic grounds, rejects the zero claim on documentary grounds, and reports which axis does the work in each case.

7 Empirical overview: 81 conflicts, 1900–2026

The contradiction-radius diagnostic requires demographic micro-data and manpower estimates, and cannot be applied sharply to every historical conflict. The 81-conflict dataset is used here only to map where civilian-share uncertainty is large, where indirect mortality dominates, and which conflicts would benefit most from identified-deaths samples; the supplement contains the per-conflict tables.

Generalisation. The diagnostic transfers to any conflict with (i) a population demographic decomposition, (ii) an independent demographic survey of the dead, (iii) an independent manpower estimate at t_0, and (iv) a reported aggregate death total. Such inputs exist for Tigray 2020–2022 (Ethiopian Statistical Service plus WHO surveys), Syria 2011– (Violations Documentation Center plus Syrian Network for Human Rights), the Mexican drug war (INEGI plus CSIS/IISS cartel estimates), the Iraq War 2003–2011 (Burnham et al., 2006), and Yemen 2014– (Yemen Data Project plus ACLED, Raleigh et al., 2010). Where these data are missing, the supplement marks the gap rather than inventing a contradiction radius.

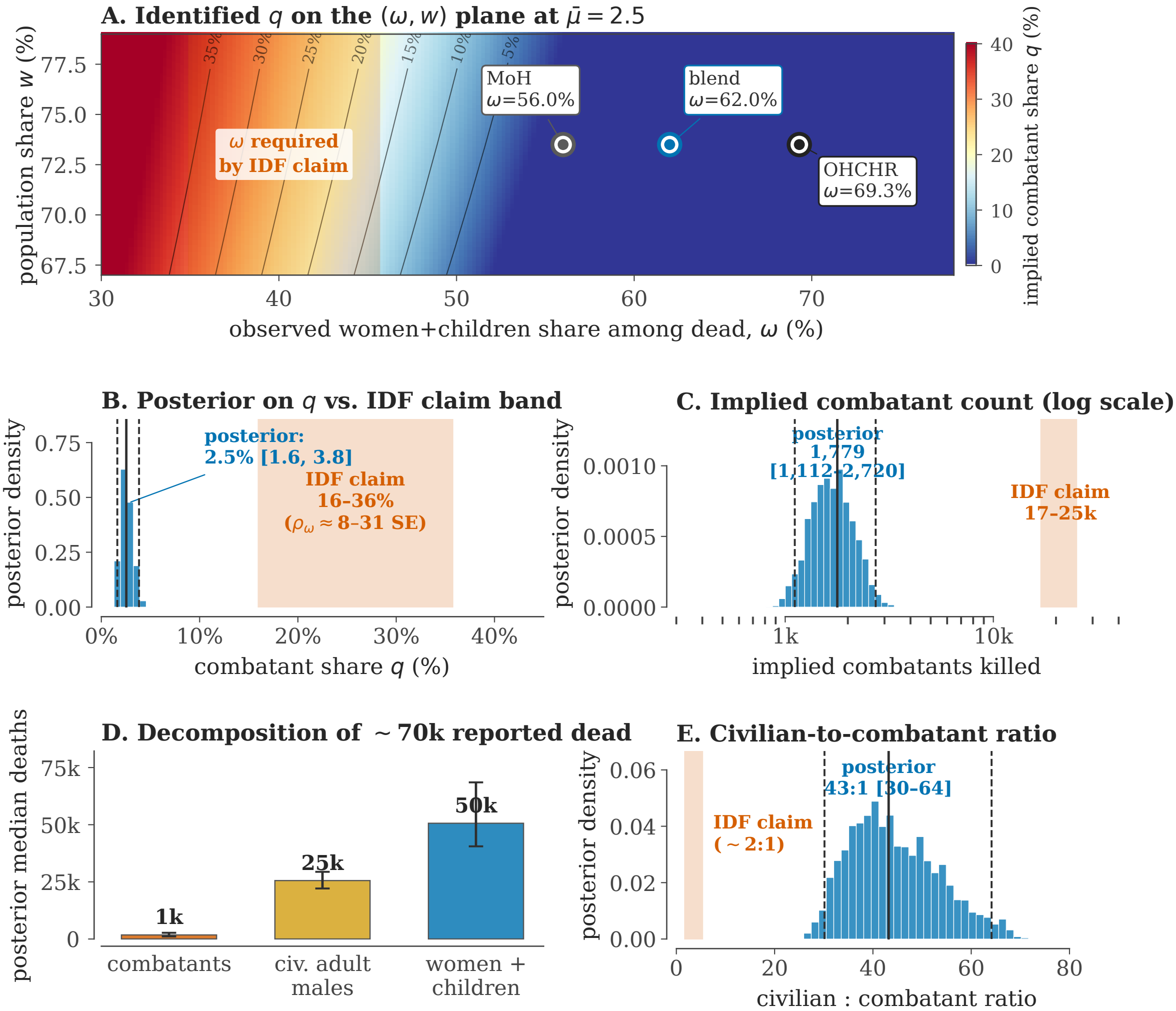

Figure 2. The diagnostic for Gaza (2023–2026). **(A)** Identified q on the (ω, w) plane at $\mu = 2.5$. Iso-q contours are black; the OHCHR, blended, and Gaza Ministry of Health (MoH) anchors all sit in the low-q (blue) region; the IDF claim requires ω in the red band on the far left, far from any observed anchor. **(B)** Histogram of the spatial posterior on q (median, 95% credible interval as solid+dashed lines) vs. the IDF claim band converted to q. **(C)** Implied combatants killed (log axis) vs. the IDF claim of 17–25k. **(D)** Posterior decomposition of the ∼70k reported dead under the spatial model: ∼1–3k combatants, ∼25k civilian adult men, ∼45–50k women and children. **(E)** Civilian-to-combatant ratio: spatial-model posterior median 43:1 vs. the IDF-implied ∼ 2:1. Panels B–E display the spatial model of Appendix B; the paper's headline claims rest on panels A and Table 4, not on the posterior.

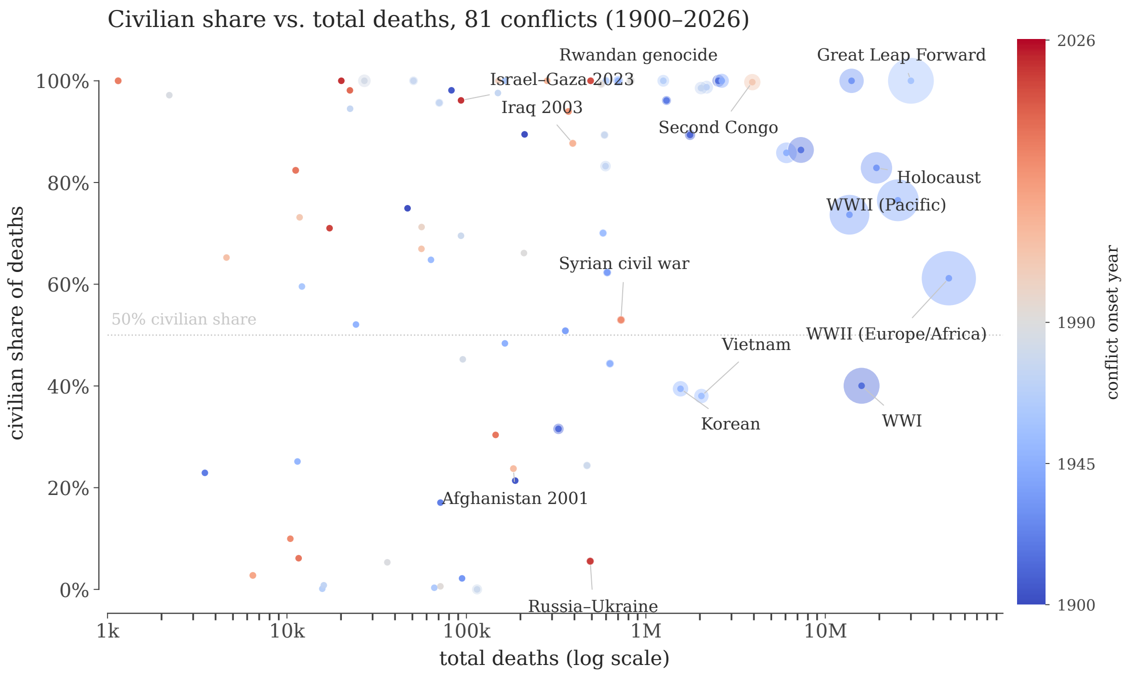

Figure 3. Civilian share vs scale, 1900–2026. Markers are conflicts; area is proportional to midpoint total deaths. Wars cluster bimodally: combat-dominated interstate wars (civilian share $\lesssim 30\%$) and civilian-targeting events, famines, and atrocities (civilian share $\gtrsim 90\%$).

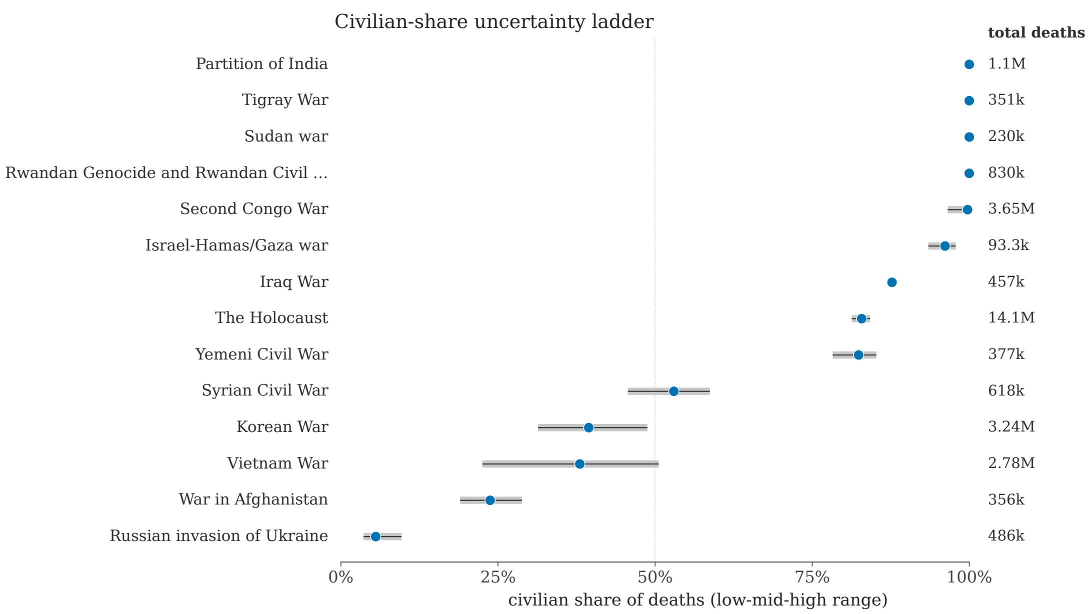

Figure 4. Civilian-share uncertainty ladder. Selected high-leverage conflicts. Grey segments are the side-level low/high implied civilian-share interval; blue dots are the midpoints. Wide intervals mark conflicts where the next technical advance is a better identified-deaths sample, not a better plot.

8 Discussion

What the paper claims, and what it does not. The conceptual contribution is the contradiction triangle (Theorem 4.3): no single number can be manipulated without forcing matching deviations in independently-measured quantities, and the contradiction radius ρ gives a unit-free measure of how far a claim sits from the consensus-feasible set. Whereas Spagat et al. (2009) frame war-death estimation as an *arena of contestation*, our framework gives the analyst a numerical referee for that arena, in the spirit of Lum et al. (2013) for multiple systems estimation and Radford et al. (2023) for Bayesian source aggregation. In the Gaza application the referee's ruling is deliberately modest: the data bound the combatant share of direct deaths at 25% with no behavioural assumptions and at about 6% under empirically calibrated exposure, and the public claim of 24–36% is rejected under every anchor and every calibrated exposure assumption we examined—its upper endpoint is arithmetically infeasible outright, and its lower endpoint survives only by asserting that civilian men were no more exposed than women and children ($\mu \approx 1$), against the calibration evidence. The paper does *not* claim that the spatial model's $q = 2.5\%$ is the truth; that number is one prior-dependent point inside the set, reported with its full model documentation so that readers can locate exactly which assumptions produce it.

Where the power comes from. It is worth being explicit about why the bound binds. The demographic relation has one free behavioural parameter, the exposure ratio μ, and the data pin the product, not the factors: a given deficit of women and children among the dead can be explained by combatants dying, by civilian men being more exposed, or by a mixture. High-combatant claims must therefore assert that $\mu \approx 1$—that civilian men in an urban war were no more exposed than women and children—at the same time as independent calibrations on conflicts with recorded combatant status put μ at 2 or higher (Frost, 2026; Cockerill, 2024). The claim is not rejected by any single measurement but by the joint configuration; that is the under-fudgibility mechanism doing its work, and it is also why the diagnostic is hard to game: an actor wishing to defeat it must coordinate distortions across a census bureau, a health ministry's death records, foreign military-balance assessments, and historical calibration data that predate the conflict.

Relation to convergent evidence. The convergence table (Table 4) is, to our knowledge, the first side-by-side placement of the demographic-decomposition literature, the named-combatant documentary record, and a generative spatial model on the same scale. We read the residual disagreement—roughly $q \in [2\%, 25\%]$—as an honest statement of what current public data cannot resolve: it is exactly the width of the exposure-calibration question plus the classification question (who counts as a combatant, on which the named-list evidence and demographic methods can legitimately differ). Narrowing it further requires either a defensible conflict-specific measurement of μ or transparent, auditable combatant lists, not another aggregation rule.

Limitations.

1. Class AM is treated as a single bucket; finer age partitions (18–29, 30–49, 50+) tighten the bounds when age-resolved data exist, and the calibration of μ from near-cutoff age bands (Frost, 2026) is itself subject to sampling noise in small historical samples.
2. The exposure range $[\mu_{\text{lo}}, \mu_{\text{hi}}]$ is imported from historical calibration; if the present conflict's exposure structure is genuinely outside that range, only the exposure-agnostic bound and the manpower bound survive. We report both throughout for this reason.

3. The radius standardises by *sampling* error; systematic error in an anchor (e.g. selection in who gets recorded as dead) is addressed by re-computing under alternative anchors, not by the radius itself. Section 6 therefore reports all anchors, with the most claim-favourable one as primary.
4. The radius is conservative for non-Gaussian errors; a bootstrap or non-parametric posterior is preferred for small demographic samples.
5. Combatant status is a legal-institutional category, not a demographic one. The framework bounds the share of deaths attributable to members of armed groups as counted by manpower estimates; disputes about who should count as a combatant move M and the interpretation of named lists, and enter the diagnostic through those channels.
6. The framework does not identify indirect deaths from blockade, famine, or healthcare collapse; these require separate excess-mortality methodology (Checchi and Roberts, 2008; Burnham et al., 2006; Degomme and Guha-Sapir, 2010; Gómez-Ugarte et al., 2025) and enter only via the denominator D, which we vary as a sensitivity dimension.

Reproducibility. All bounds, radii, and table entries in this paper are generated by a validation script that recomputes them from the raw inputs and fails on any discrepancy. In addition, the algebraic core of the framework (Appendix A) is formalised and machine-checked in Lean 4 with mathlib, and the remaining results are verified symbolically and numerically by dedicated scripts. Code, the formal proofs, the spatial simulator, and all inputs are available in the companion repository.

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A Proofs

The mathematical core of the framework has been formalised and machine-checked in Lean 4 with mathlib: the benchmark identity (2), the inversion, monotonicity, and derivative of $q(\mu)$ in Theorem 3.3, the η-slack bias identities and exact bias values of Remark 3, the manpower bound of Corollary 3.4, the delta-method gradients, the Gauss–Markov optimality of Proposition 2.1, the monotonicity and infimum facts underlying Theorem 4.3, the two-point contamination kernel of Proposition 5.1 (mean displacement εR versus full-R quantile movement), and the Section 6 grid arithmetic over exact rationals. The development compiles with no unproven statements and is in the `formal/` directory of the companion repository; the remaining measure-theoretic steps are verified symbolically and by Monte-Carlo stress tests in the repository's verification scripts. The human-readable proofs follow.

Proof of Theorem 3.3. Let λ be the civilian per-capita death hazard in the women-and-children class W, so class AM civilians have hazard $\mu\lambda$. Expected civilian deaths are proportional to $\lambda wN + \lambda\mu(a-f)N$, so the civilian share landing in W is $w/(w+\mu(a-f))$. Since only the civilian fraction $1-q$ of deaths reaches W,

$$\omega = (1-q)\,\frac{w}{w+\mu(a-f)},$$

which solves to (3); moreover $\partial_\mu q = -\omega(a-f)/w \le 0$, so $q(\mu)$ is monotone decreasing. Sharpness: every $\mu \in [\mu_{\mathrm{lo}}, \mu_{\mathrm{hi}}]$ with $q(\mu) \in (0,1)$ corresponds to a feasible hazard pair $(\lambda, \mu\lambda)$, $\lambda > 0$, that attains $q(\mu)$ exactly; the boundary values $q(\mu) \in \{0,1\}$ are attained in the closure (e.g. $q = 1$ means $\omega = 0$, no civilian deaths, the limit $\lambda \downarrow 0$). Hence no sub-interval can be excluded. The benchmark identity (2) is the case $\mu = 1$, using $w + a = 1$. Corollary 3.4 follows since $D_M \le M$ implies $q = D_M/D \le M/D$. □

Proof of Proposition 2.1. Among linear unbiased combinations $\sum_i c_i\hat{\theta}_i$ with $\sum_i c_i = 1$, the variance $\sum_i c_i^2\hat{\sigma}_i^2$ is minimised by $c_i \propto 1/\hat{\sigma}_i^2$ (Gauss–Markov / Lagrange). The bias expression follows by linearity of expectation. For the contamination statement, source i contributes weight $\hat{\sigma}^2_{\mathrm{agg}}/\hat{\sigma}_i^2$ to the aggregate; with probability ε_i its report is replaced by a draw from a distribution supported on a set of diameter R_i, displacing the aggregate by at most $\varepsilon_i R_i\,\hat{\sigma}^2_{\mathrm{agg}}/\hat{\sigma}_i^2$ in expectation, and summing over sources gives the bound. □

Proof of Theorem 4.3. ($\Leftarrow$) If $q^{(i)} \in \mathcal{F}(\hat{x})$, the point $(\hat{\omega}, \hat{w}, \hat{M})$ attains objective zero. ($\Rightarrow$) Suppose $\rho^{(i)} = 0$ and take feasible (ω'_n, w'_n, M'_n) with objective $\to 0$. Because the M-penalty is one-sided, this yields $\omega'_n \to \hat{\omega}$, $w'_n \to \hat{w}$, and $(M'_n - \hat{M})_+ \to 0$ only (not $M'_n \to \hat{M}$). Each feasible point has some $\mu_n \in [\mu_{\mathrm{lo}}, \mu_{\mathrm{hi}}]$ with $q^{(i)} = 1 - \omega'_n(w'_n + \mu_n(a-f))/w'_n$; passing to a convergent subsequence $\mu_n \to \mu^*$ in the compact interval and using continuity gives $q^{(i)} = 1 - \hat{\omega}(\hat{w} + \mu^*(a-f))/\hat{w}$. For the manpower constraint, $q^{(i)}D \le M'_n \le \hat{M} + (M'_n - \hat{M})_+ \to \hat{M}$, so $q^{(i)}D \le \hat{M}$. Hence $q^{(i)} \in \mathcal{F}(\hat{x})$. For the second statement, any rationalisation (ω', w', M') satisfies, by definition of the infimum, $\max\{|\omega' - \hat{\omega}|/\sigma_\omega,\ |w' - \hat{w}|/\sigma_w,\ (M' - \hat{M})_+/\sigma_M\} \ge \rho^{(i)}$, so at least one penalised displacement is $\ge \rho^{(i)}$ standardised units. □

Proof of Proposition 5.1. (i) Condition on the realised pattern T. The sources outside T are honest, so any posterior formed from them plus arbitrary reports from the sources in T is supported on the identified set with the T-sources removed, which has diameter R_T and contains the face-value posterior support (in particular both L_0 and U_0); hence each endpoint of any support-respecting credible interval lies within R_T of the corresponding endpoint of I_0.

(ii) The pattern is T with probability $P(T) = \prod_{i\in T}\varepsilon_i \prod_{i\notin T}(1-\varepsilon_i)$, so the expected endpoint displacement is at most $\sum_T P(T)R_T$. Splitting off the singleton patterns, $\sum_T P(T)R_T \le \sum_i \varepsilon_i R_i + \sum_{|T|\ge 2} P(T)R_T$, and $P(T) = O(\varepsilon^{|T|})$ makes the second sum higher-order; it vanishes when at most one source can be contaminated. Note that R_T for $|T| \ge 2$ is *not* bounded by $\sum_{i\in T} R_i$ in general (two mutually corroborating sources can each have a small removal diameter while their joint removal is much larger), which is why (6) is stated as a first-order band rather than a uniform envelope.

(iii) By assumption the posterior places weight at most τ on contaminated components, so the contaminated and face-value posterior CDFs differ by at most τ uniformly. With density

$\geq m$ on the segment between $\hat{Q}_0(u)$ and $\hat{Q}_\varepsilon(u)$, the mean value theorem converts the CDF gap into the quantile gap τ/m. The weight condition is substantive: posterior mixture weights are proportional to $P(T)$ times the component marginal likelihoods, so an adversarial component that fits the observed data *better* than the clean model can carry posterior weight exceeding its prior probability τ. $\square$

B The spatial Bayesian model

This appendix documents the forward model, priors, likelihood, and inference behind the posterior reported in Section 6, Step 5. Code and inputs are in the `gaza_sim/` directory of the companion repository, and the posterior is reproducible from a fixed random seed. We emphasise its role: it is *one* model-based point within the assumption-free identified set, and its low posterior q reflects its priors and structural choices as much as the data. Readers who reject those choices should fall back to the bounds, which do not depend on them.

Geometry and population. Gaza's five governorates are split into $C = 400$ equal-population cells (80 per governorate), with cell populations N_c from the PCBS 2023 total. Demographic fractions (women-and-children share w, adult-male share a) are held constant across cells at the Gaza-wide values.

Latent parameters and priors. Each posterior draw samples, independently and uniformly on the stated ranges:

Parameter	Range	Meaning
M	$[35\text{k}, 60\text{k}]$	total militant stock
γ	$[0, 1.5]$	spatial clustering amplification of militants
σ	$[0.3, 1.2]$	SD of the cell-level latent log-density
α	$[0, 8]$	strike-targeting concentration on militant density
β	$[0.6, 1.6]$	targeting non-linearity exponent
μ_M	$[1, 2.5]$	civilian adult-male exposure multiplier
ϵ_C	$[0.7, 1]$	child exposure down-weight
$\bar{d}$	$[1.2, 2.8]$	expected kills per strike
K	$[29\text{k}, 40\text{k}]$	total air-to-ground strikes
u	$[1/1.7, 1]$	recovery factor (MoH observed / true direct)

Militants are allocated across cells proportionally to $N_c \exp(\gamma u_c)$ with $u_c \sim \mathcal{N}(0, \sigma^2)$, capped at 30% of each cell's adult-male population and rescaled to the total M.

Strike-kill model. Strikes are allocated across cells with weight $N_c(1 + \alpha m_c/N_c)^\beta$, where m_c is the militant count in cell c. Within a cell, a death is a militant, a civilian adult man, or a woman/child with probabilities proportional to m_c, $(N_c a - m_c)\mu_M$, and $N_c w\, \epsilon_C$ respectively. Expected class-specific deaths are summed over cells; observed deaths are true direct deaths times the recovery factor u.

Likelihood. Three observables: (i) the MoH cumulative total (log-Gaussian, 7% scale); (ii) the adult-male share among the dead against the OHCHR verified sample (Beta likelihood at its sample size, $n = 8{,}119$); and (iii) the same share against the MoH full record, down-weighted to an effective sample size of $n/5$ to reflect cruder demographic categorisation. The two demographic likelihoods enter with weight 1/2 each.

Inference. Importance sampling with 2×10^5 prior draws; the effective sample size and all posterior quantiles are computed under the normalised weights. The reported posterior is $q = 2.5\%$ (95% credible interval $[1.6\%, 3.8\%]$).

Sensitivities and limitations. The posterior concentrates at low q for two structural reasons that a critic can legitimately contest: the exposure multiplier prior is capped at $\mu_M \leq 2.5$ (values above the cap would push q lower still, but a mass of prior near $\mu_M = 1$ pulls the posterior up much less than the OHCHR-weighted demographic likelihood pulls it down), and the OHCHR sample—which over-represents residential-strike deaths where women and children predominate (Spagat and Cockerill, 2024)—enters the likelihood at full sample weight. Re-anchoring the demographic likelihood entirely on the MoH record moves the posterior median up but keeps it far inside the identified set. Because these choices are contestable, the paper's claims do not rest on this posterior; it appears as one row of Table 4.

C The 81-conflict dataset

Table 5 lists every conflict in the dataset behind the overview of Section 7, with midpoint death totals and the implied civilian-share intervals. The table is generated by the same script that draws Figures 3 and 4; per-conflict source notes, the indirect-death decomposition, and the accounting rules are documented in the supplement.

Table 5. The 81-conflict dataset behind Figures 3 and 4 (conflicts with at least 1,000 attributed deaths shown). Deaths are midpoints of the source ranges (direct military + civilian, including indirect where sources attribute it); the civilian-share interval spans the most and least civilian-heavy readings of the source ranges. Per-conflict sources are in the supplement.

Conflict	Period	Deaths (mid)	Civilian share	lo–hi
Boxer Rebellion	1899–1901	47k	75%	63–92%
Philippine-American War	1899–1902	211k	89%	86–92%
Herero and Namaqua Genocide	1904–1908	82.5k	98%	97–99%
Russo-Japanese War	1904–1905	187k	21%	19–24%
Mexican Revolution	1910–1920	1.76M	89%	73–95%
First and Second Balkan Wars	1912–1913	326k	32%	18–43%
Armenian Genocide and late Ottoman genocide...	1914–1923	2.54M	100%	100–100%
World War I	1914–1918	15.9M	40%	31–47%
Russian Civil War	1917–1923	7.32M	86%	41–97%
Greco-Turkish War and population exchange	1919–1922	1.3M	96%	94–98%
Irish War of Independence and Civil War	1919–1923	3.49k	23%	19–27%
Rif War	1921–1927	71.6k	17%	11–31%
Chaco War	1932–1935	94.5k	2%	1–4%
Soviet repression: Holodomor and Great Purge	1932–1939	14M	100%	100–100%
The Holocaust	1933–1945	19.3M	83%	81–84%
Second Italo-Ethiopian War	1935–1937	609k	62%	49–84%
Spanish Civil War	1936–1939	356k	51%	38–61%
Second Sino-Japanese War	1937–1945	13.6M	74%	62–83%
World War II - European/African/Atlantic Th...	1939–1945	48.8M	61%	56–66%
World War II - Pacific Theater/Asia-Pacific...	1941–1945	25.4M	77%	67–83%
Bengal Famine of 1943	1943–1944	2.65M	100%	100–100%
Chinese Civil War	1945–1949	6.07M	86%	86–86%
Indonesian National Revolution	1945–1949	164k	48%	19–72%
First Indochina War	1946–1954	631k	44%	27–60%
1948 Arab-Israeli War/Nakba	1947–1949	24.2k	52%	42–62%
Partition of India	1947–1948	700k	100%	100–100%
Malayan Emergency	1948–1960	11.4k	25%	22–28%
Korean War	1950–1953	1.56M	39%	31–49%
Mau Mau Uprising	1952–1960	63.4k	65%	45–83%

Table 5 continued

Conflict	Period	Deaths (mid)	Civilian share	lo–hi
Algerian War of Independence	1954–1962	577k	70%	63–75%
Vietnam War	1955–1975	2.04M	38%	23–51%
Great Leap Forward famine	1958–1962	30M	100%	100–100%
Congo Crisis	1960–1965	12.1k	60%	59–60%
Guatemalan Civil War	1960–1996	164k	100%	100–100%
North Yemen Civil War	1962–1970	66.2k	0%	0–0%
Indonesian mass killings of 1965-66	1965–1966	600k	100%	100–100%
Nigerian Civil War	1967–1970	1.25M	100%	100–100%
Six-Day War	1967–1967	15.7k	0%	0–0%
Bangladesh Liberation War/1971 genocide	1971–1971	2.18M	99%	89–100%
Yom Kippur War	1973–1973	16k	1%	0–2%
Ethiopian Civil War/Derg/1983-85 famine	1974–1991	596k	83%	81–85%
Angolan Civil War	1975–2002	70.6k	96%	95–96%
Indonesian occupation of East Timor	1975–1999	150k	98%	96–98%
Khmer Rouge Cambodian Genocide	1975–1979	2.03M	99%	98–99%
Lebanese Civil War	1975–1990	22.5k	95%	88–97%
Mozambican Civil War	1977–1992	50.7k	100%	100–100%
Cambodian-Vietnamese War	1978–1989	588k	89%	83–94%
Salvadoran Civil War	1979–1992	93.3k	70%	61–77%
Soviet-Afghan War	1979–1989	115k	0%	0–0%
Iran-Iraq War	1980–1988	470k	24%	17–33%
Second Sudanese Civil War	1983–2005	27k	100%	100–100%
Sri Lankan Civil War	1983–2009	95.5k	45%	42–48%
First Intifada	1987–1993	2.21k	97%	97–97%
Nagorno-Karabakh wars	1988–2023	36.2k	5%	4–7%
Gulf War	1990–1991	209k	66%	41–90%
Rwandan Genocide and Rwandan Civil War	1990–1994	822k	100%	100–100%
Somali Civil War	1991–	563k	99%	99–100%
Yugoslav Wars - Bosnia/Croatia	1991–1995	71.6k	1%	0–1%
First Chechen War	1994–1996	56.3k	71%	56–86%
First Congo War	1996–1997	151k	100%	100–100%
Kosovo War	1998–1999	11.8k	73%	71–75%
Second Congo War	1998–2003	3.92M	100%	97–100%
Second Chechen War	1999–2009	56.2k	67%	48–82%
Second Intifada	2000–2005	4.6k	65%	53–75%
War in Afghanistan	2001–2021	183k	24%	19–29%
Darfur conflict/genocide	2003–2020	282k	100%	100–100%
Iraq War	2003–2011	391k	88%	87–88%
Mexican Drug War	2006–	6.45k	3%	2–3%
Boko Haram insurgency/Lake Chad conflict	2009–	370k	94%	94–94%
Libyan Civil Wars	2011–2020	10.4k	10%	7–12%
Syrian Civil War	2011–2024	727k	53%	46–59%
South Sudanese Civil War	2013–2020	1.15k	100%	100–100%
War against ISIS in Iraq and Syria	2014–2019	145k	30%	16–47%
War in Donbas	2014–2022	11.6k	6%	6–7%
Yemeni Civil War	2014–	11.2k	82%	78–85%
Rohingya genocide/Myanmar military operatio...	2016–	22.4k	98%	96–99%
Tigray War	2020–2022	492k	100%	100–100%
Myanmar civil war	2021–	17.3k	71%	71–71%
Russian invasion of Ukraine	2022–	490k	6%	4–10%
Israel-Hamas/Gaza war	2023–	93.3k	96%	93–98%
Sudan war	2023–	20.1k	100%	100–100%